

- A lattice-free elementary plastic-deformation unit (OSTZ) is proposed.
- Dislocations emerge as a collective, large-N limit of OSTZ chains.
- Peierls stress, core energy, and stacking-fault energy are derived, not fitted.
- Validated against grain-boundary-sliding data across metals, glasses, ceramics.
- Unifies crystal and lattice-free plasticity within Eshelby's inclusion theory.